SMART STUDY RECOMMENDER USING MACHINE LEARNING

Course: Fundamentals of AI and ML

Student Name: Aditya Chouksey

Registration No.: 25BAI11531

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Artificial Intelligence (AI) and Machine Learning (ML) have become important technologies for analyzing data and making predictions. These technologies are widely used in many fields such as healthcare, finance, transportation, and education. In the education sector, AI can help analyze student performance and provide useful insights that support better learning outcomes.

Students often study multiple subjects at the same time, but it is not always easy for them to identify which subject requires more attention. Sometimes students spend time revising subjects they already understand well, while weaker subjects remain neglected. By analyzing academic performance data, machine learning can help identify these weaker areas.

This project aims to build a simple machine learning system that analyzes a student's scores in Physics, Chemistry, and Mathematics and recommends the subject that needs the most improvement. The system uses a classification model to analyze score patterns and identify the weakest subject automatically.

The goal of this project is not only to build a working machine learning model but also to demonstrate how concepts learned in the AI/ML course can be applied to solve a small real-world educational problem.

PROBLEM STATEMENT

Students often face difficulty deciding how to distribute their study time among different subjects. Without proper analysis of their performance, they may focus on subjects they are already comfortable with and overlook areas that require improvement.

Teachers and students usually rely on manual observation of marks to determine weak subjects. However, when dealing with large amounts of performance data, this process can become inefficient.

The objective of this project is to develop a machine learning model that analyzes student score data and automatically recommends the subject that requires the most attention. Such a system can help students make better study decisions and improve their academic performance.

Objectives

The main objectives of this project are:

- To apply machine learning concepts learned during the course.
- To build a simple predictive model using student score data.
- To automatically identify the weakest subject for a student.
- To generate recommendations that help students focus their study efforts.
- To demonstrate the complete workflow of a basic machine learning project, from data preparation to prediction.

DATASET

The dataset used in this project contains student scores in three core subjects:

Physics

Chemistry

Mathematics

Each row in the dataset represents the academic performance of one student. The model analyzes these scores and determines which subject has the lowest score, which is considered the subject requiring the most improvement.

Example Dataset Format

Physics	Chemistry	Maths
45	78	52
82	60	88
55	50	48

To make the model more reliable, the dataset was expanded with multiple records representing different combinations of student scores. A larger dataset helps the machine learning algorithm learn patterns more effectively and produce better predictions.

METHODOLOGY

This project follows the standard workflow used in many machine learning applications.

1. Data Collection

A dataset containing student scores in Physics, Chemistry, and Mathematics was created. These values simulate real academic performance data.

2. Data Preprocessing

Before training the model, the dataset was cleaned and organized to ensure that it could be processed correctly by the machine learning algorithm. This step helps remove inconsistencies and prepares the data for analysis.

3. Label Generation

The weakest subject for each record was identified by comparing the three scores. The subject with the lowest score was assigned as the label for that data entry.

4. Model Training

A Decision Tree Classifier was used to train the model. Decision trees are commonly used for classification tasks because they are easy to understand and interpret.

5. Model Evaluation

The dataset was divided into training and testing data. The trained model was evaluated using test data to measure its accuracy and reliability.

6. Prediction

Once the model was trained successfully, it was used to predict which subject a student should focus on based on their input scores.

TOOLS AND TECHNOLOGIES

The following tools and technologies were used during the development of this project:

- Python – Main programming language used for the project
- Pandas – Used for handling and processing the dataset
- Scikit-learn – Used for implementing the machine learning model
- Joblib – Used to save the trained model for reuse
- Matplotlib – Used to visualize the dataset and relationships between scores
- Streamlit – Used to create a simple web interface for user interaction
- GitHub – Used for version control and project hosting

These tools helped in developing the project efficiently and organizing the workflow properly.

IMPLEMENTATION

The implementation of the project consists of several main components.

1. Model Training Script

A Python script was written to load the dataset, preprocess the data, train the Decision Tree classifier, and save the trained model using Joblib.

2. Data Visualization

Graphs were created using Matplotlib to better understand the relationships between subject scores. Visualization helps in identifying patterns and gaining insights from the dataset.

3. Web Application Interface

A simple Streamlit web application was developed to allow users to interact with the system. Students can enter their scores in Physics, Chemistry, and Mathematics through the interface.

After the scores are entered, the system uses the trained machine learning model to predict the subject that requires the most improvement.

The trained model is stored as a .pkl file, which allows the application to reuse the model without retraining it every time the system runs.

RESULTS

The developed system successfully analyzes the input scores and predicts the subject that a student should focus on.

Example

Input Scores

Physics: 40

Chemistry: 75

Mathematics: 50

Model Output

Recommended Subject: Physics

This result indicates that Physics is the weakest subject based on the provided scores, and the student should focus more on improving it.

Although the system is simple, it demonstrates how machine learning can assist in educational decision-making.

Challenges

During the development of this project, several challenges were encountered:

• Designing a dataset that represents realistic student performance

- Understanding how classification algorithms work in practice
- Avoiding bias in the dataset that could affect model predictions
- Integrating the trained model with the Streamlit web interface

Overcoming these challenges helped improve my understanding of how machine learning systems are developed and deployed.

FUTURE IMPROVEMENTS

There are several ways this project can be expanded and improved in the future:

- Use a larger dataset based on real student performance data
- Provide topic-level recommendations instead of only subject-level suggestions
- Improve prediction accuracy using advanced machine learning algorithms such as Random Forest or Gradient Boosting
- Develop a full web application or mobile application for easier access
- Add features such as study recommendations, progress tracking, or performance analytics

These improvements could make the system more practical and useful for real educational environments.

CONCLUSION

This project demonstrates a basic application of machine learning in the field of education. By analyzing student scores, the system can identify weak subjects and recommend where students should focus their study efforts.

Although the model used in this project is relatively simple, it effectively illustrates how AI concepts such as data preprocessing, model training, prediction, and evaluation can be applied to solve practical problems.

Working on this project provided valuable experience in building a complete machine learning pipeline and integrating it with a user-friendly interface. It also showed how AI technologies can support students in improving their learning strategies.

Learning Outcomes

Through this project, the following concepts and skills were learned:

- Understanding the machine learning workflow
- Data preprocessing and dataset preparation
- Supervised learning and classification techniques
- Model evaluation and prediction methods
- Building a simple AI application with a web interface
- Using GitHub for version control and project management

Overall, this project helped strengthen my understanding of AI and machine learning concepts while demonstrating their practical applications.